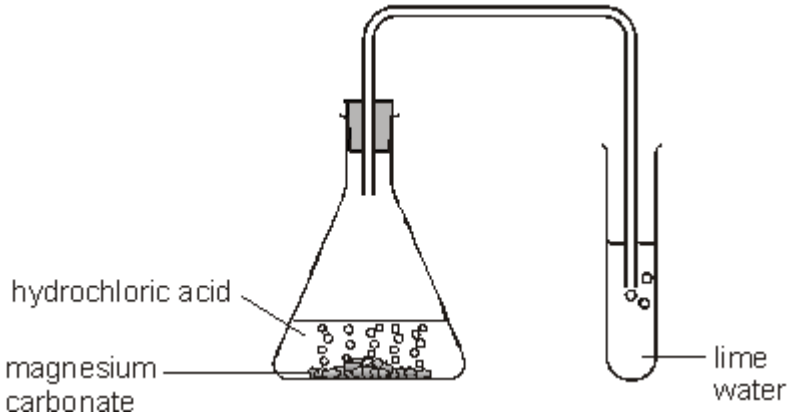
Q1.

The word equation for the reaction between magnesium carbonate and hydrochloric acid is shown below.

magnesium carbonate + hydrochloric acid → magnesium chloride + carbon dioxide + water

(a) Sadiq added hydrochloric acid to magnesium carbonate in a flask.



(i) Suggest the pH of hydrochloric acid.

.....

(ii) The carbon dioxide produced was bubbled through lime water.

How would the lime water change?

.....

2 marks

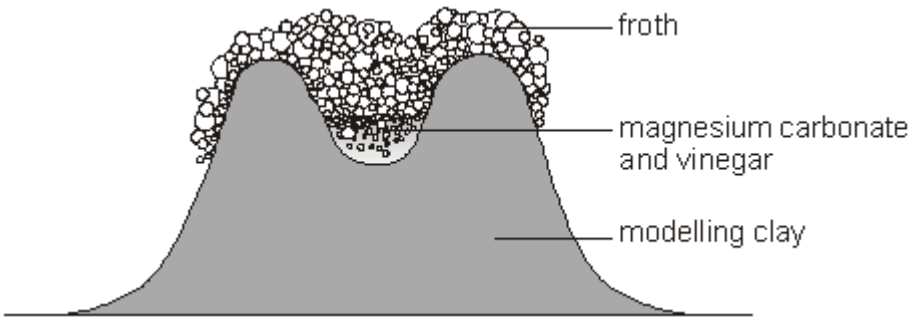
(b) Sadiq repeated the experiment by adding **sulphuric acid** to magnesium carbonate.

Complete the word equation for the reaction that took place.

magnesium carbonate + sulphuric acid → ..... + carbon dioxide + water

1 mark

(c) Sadiq made a model volcano.
He put magnesium carbonate into the model.
He added vinegar and a drop of washing-up liquid.



The mixture fizzed, and froth poured out of the model volcano.

(i) The vinegar reacted with the magnesium carbonate.

Suggest the pH of vinegar.

.....

(ii) The froth running down the side of the model represents part of a real volcano.
Give the name of this part.

.....

2 marks
maximum 5 marks

Q1.

(a) (i) • a number less than 7

1 (L5)

(ii) • it would turn milky **or** cloudy

*accept 'white **or** chalky'*

1 (L6)

(b) • magnesium sulphate

1 (L6)

(c) (i) • a number less than 7

1 (L5)

(ii) • lava

*do **not** accept 'magma'*

1 (L5)

[5]